TERM 4

CHAPTER 12

E-COMMUNICATIONS

CHAPTER OVERVIEW	
Unit 12.1	Register a web-based email address
Unit 12.2	Managing email

By the end of this chapter, you will be able to:

- Describe how to manage email inboxes in relation to organising by using folders or labels
- Prioritise emails in your inbox
- Create a distribution list
- Register a web-based email address

INTRODUCTION

An electronic communication device refers to any type of computerised device (instrument, equipment or machine) or software that can compose, read or send any electronic message using radio, optical or other electromagnetic systems. An electronic message can be a text message, electronic mail, an instant message like WhatsApp, teleconferencing, social networking, Skype, blogs or even access to an internet site.

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UNIT

12.1 Register a web-based email address

TIPS FOR OPENING A GMAIL ACCOUNT

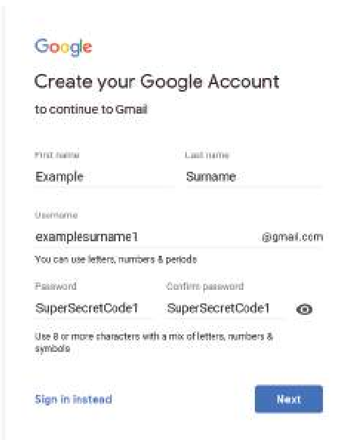
Learn how to create a Gmail account

https://support.google.com/

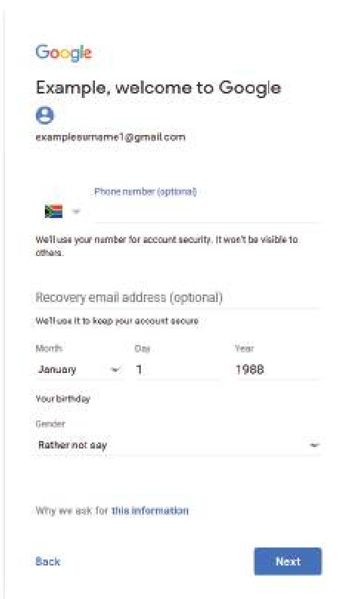
Having an email address is a must in this day and age. You will use your email address for many online activities, such as signing up for a social media account or filling out online applications. Luckily, creating an email account using a web-based service is relatively easy. In this section, you will learn how to create a Gmail account.

Guided Activity 12.1

1. Use your web browser to go to the Gmail website (www.gmail.com).
2. Click on the *Create an Account* button. A new page should now open.
3. Enter your name and surname.
4. Choose a username for your account.
5. Enter a password for your account and enter the same password in the *Confirm password* space.
6. Click the *Next* button.



7. Pay careful attention to the items on the sign-up form that are optional. You can skip these items if you want to. For Google, submitting a phone number or recovery email address allows Google to help you recover your password if you forget it.
8. Fill in your birthday and gender before clicking next.



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Guided Activity 12.1

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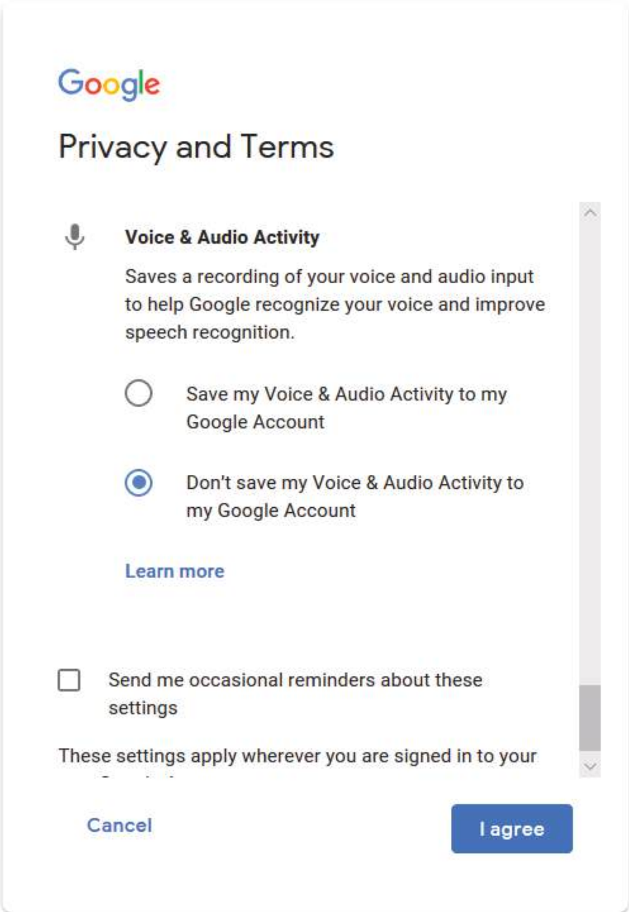
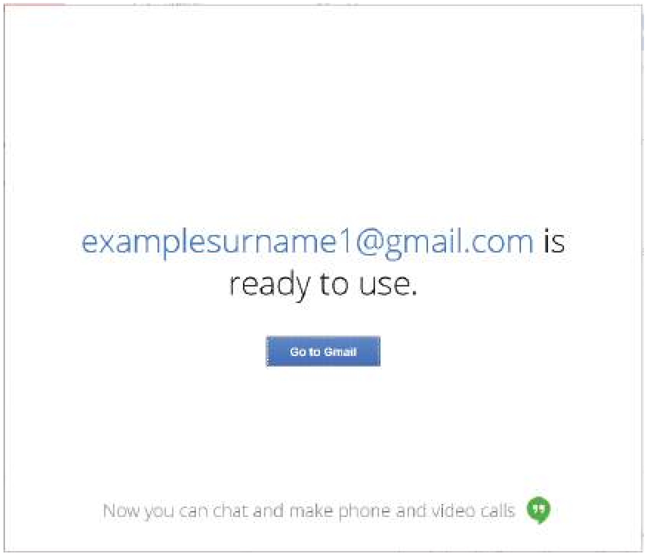
9. Read through Google's *Privacy and Terms* agreement.
10. Read through the custom options and select which information Google is allowed to record about you.
11. Once you have selected options you are comfortable with, click *I agree*.



12. The moment you click *I agree*, Google will create your new email account for you.



DO'S AND DONT'S FOR WRITING AN EMAIL

Learn about the Do's and Dont's of writing an email

Bowvalleycollege.libguides.com

DO'S AND DON'TS WHEN WRITING AN EMAIL

There are certain "rules" to follow when you are writing an email, such as:

- Always make sure that your subject line is a clean and brief indication (summary) of what the email is about.
- Use a more professional **salutation**. If you do not know the person you are writing to, start your email with "Dear (insert name)". If the situation is not as formal, you can use "Hi" or "Hello" instead. NEVER use any slang form of greeting such as "Yo" or "Howzit" in an email.
- Do not try to be funny when writing an email. It can easily be misunderstood and seldom works out the way you want it to. Remember, the recipient cannot see your facial expressions.
- Always proofread your message before you send it. Use your spellchecker to make sure that there are no spelling mistakes or typos. People are judged by the way their emails look.
- Do not assume that the recipient knows what you are talking about. Even if you think that they should know, make sure that your email can be read and understood without pre-knowledge. It can be very frustrating to look back through a whole string of emails to try and find something that was said previously.
- Always reply to an email. Even if you do not have the answer right away, let the recipient know that you have received their mail and will get back to them.
- Do not overuse exclamation marks. People interpret this as aggressive behaviour and seldom respond positively.
- Never send an email when you are angry. First think about what you want to say and how you want to say it. You can even save the email in the Drafts folder and go back to it after you have cooled down. Bad emails can never be retracted.
- Do not copy the whole world in your email. Send it only to the people who would really be interested in it.
- Do not play email ping-pong. If you have to reply more than twice to the same topic, rather pick up the phone and call the person.
- Do not send large attachments. Rather send a link to download the file or compress it before attaching it.

Activity 12.1

Using a computer in class:

1. Create a Gmail account.
2. Create a distribution list.
3. Organise your emails by creating labels.
4. Set up filters in Gmail.
5. Apply the Gmail star system to your emails.
6. Prioritise your emails.

Answer the following in your own words:

1. Describe how you create a Gmail account.
2. Give five examples of good email experiences you have had. Give a reason for why you found each experience good.
3. Give five examples of bad email experiences you have had. Give a reason for why you found each experience bad.
4. Based on your experiences, make your own list of five email do's and don'ts.

UNIT

12.2 Managing email

Dealing with emails is one of the most time-consuming tasks in a work environment. In 2018, it was estimated that dealing with emails was taking up almost 11 hours of the work week.

In this chapter, you will learn how to work with Gmail, but the process is the same for most other web-based email applications, such as Outlook.com or Hotmail.com.

ORGANISE USING LABELS

Gmail is a label-based system where the labels are the same as folders.

Labels are a great way to organise Gmail. They are like tags you can add to emails you send or receive. Unlike folders, you can add more than one label to an email. You can also colour-code these labels to make it easier to see into which category an email falls.

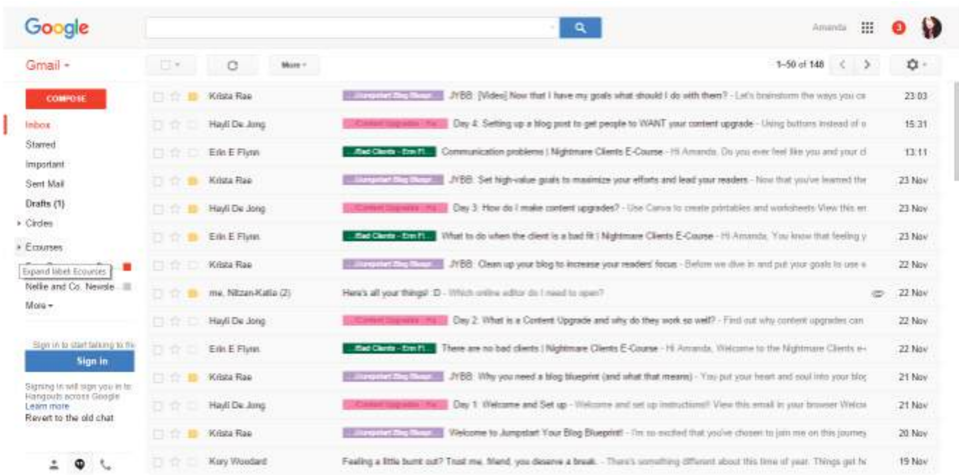


Figure 12.1: *Emails can be colour-coded according to their labels*

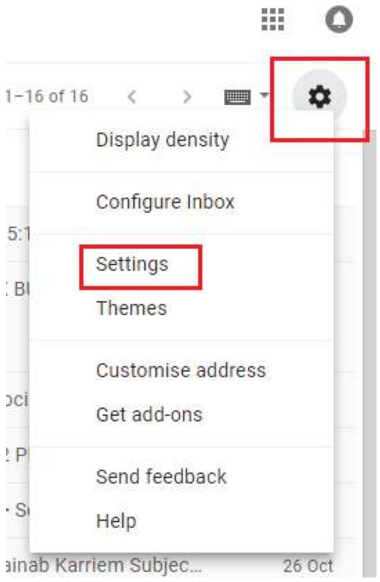
It is important to keep your system as easy and simple as possible. Do not have too many labels or categories. For example, you can create labels for:

- family
- friends
- school
- activities.

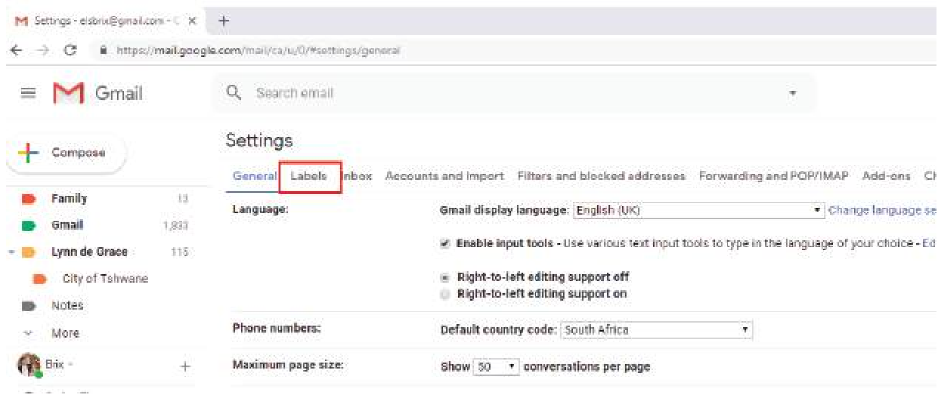
Also remember, you can add more than one label to your emails. An email coming from your school about an upcoming swimming event can be labelled both *School* and *Activities*.

Guided Activity 12.2

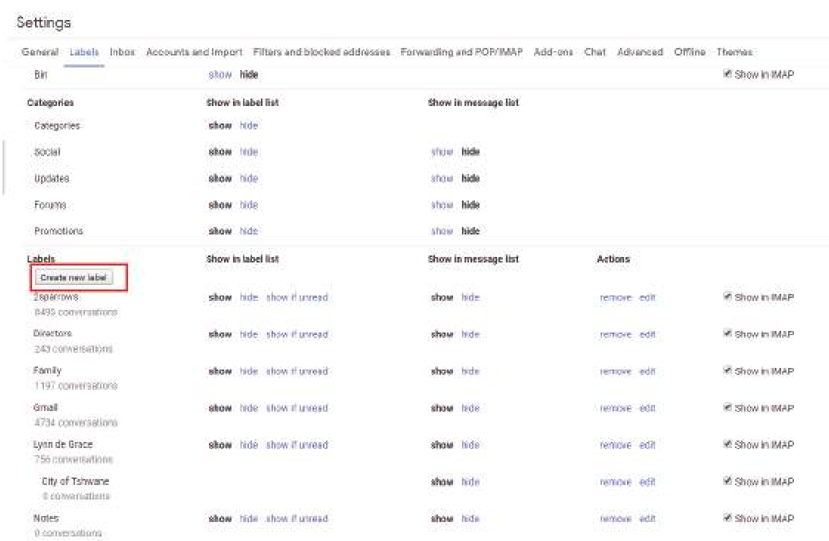
1. Open your Gmail account.
2. Click the *Settings* gear icon near the top right corner of the Gmail screen.



3. Follow the *Settings* link in the menu that comes up.



4. Go to the *Labels* tab shown above.
5. Click *Create new label* in the *Labels* section.



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Guided Activity 12.2

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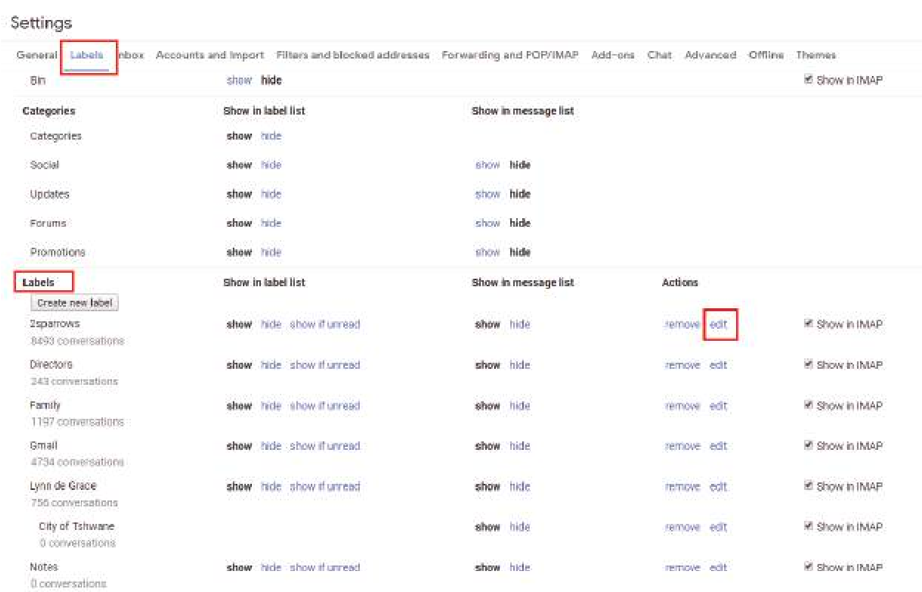
6. Type the new label's desired name under *Please enter a new label name.*

7. Check *Nest label under*: and select a label from the drop-down menu.

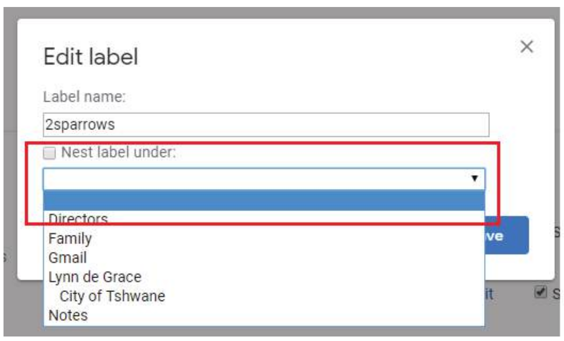
Guided Activity 12.3

To move an existing label beneath another label:

1. In the section of the *Labels* tab, click *Edit* in the *Actions* column for the label you want to move.



2. Check *Nest label under* and select a destination from the drop-down menu.

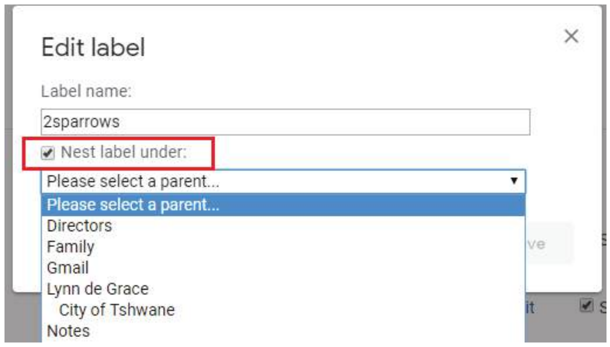


3. Click *Create* or *Save*.

Guided Activity 12.4

To move a Gmail folder to the top or turn it into a subfolder:

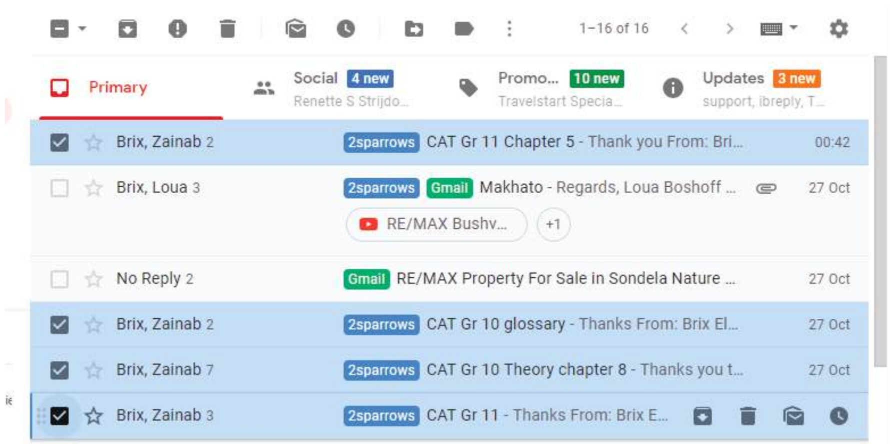
1. In the section of the *Labels* tab, click *Edit* in the *Actions* column for the label you want to move.
2. To move the label beneath another label:
   a. Make sure *Nest label under:* is checked.
   b. Select the label under which you want to move the label from the drop-down menu.



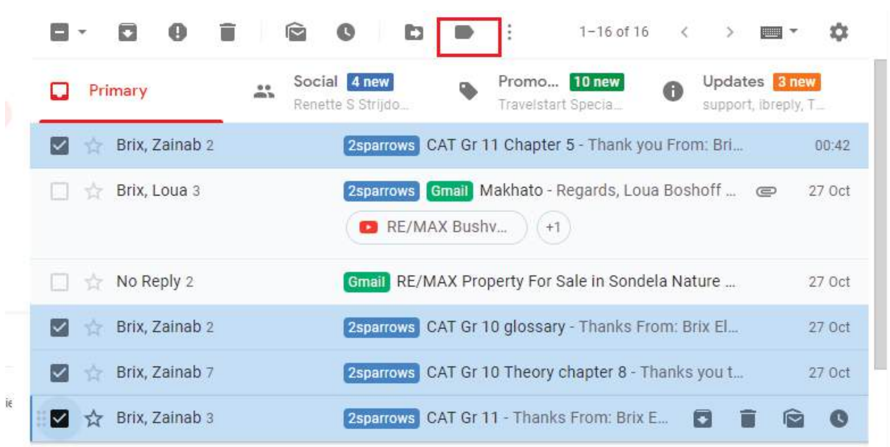
3. To move the label to the top, make sure *Nest label under:* is not checked.
4. Click *Save*.

Guided Activity 12.5

1. Select each message by checking the box to the left of each message you want to label.



2. Click the *Labels* icon in the menu.



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Guided Activity 12.5

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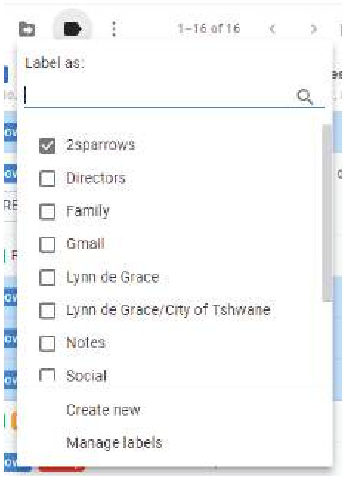
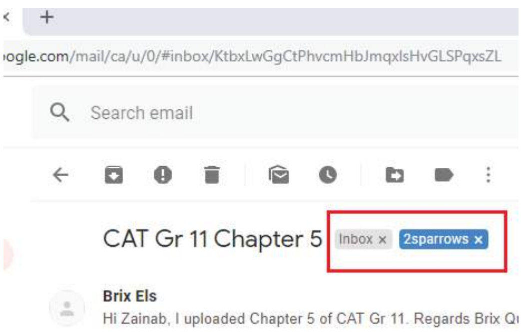
3. Check the boxes next to each label you want to assign to your selected messages or click *Create new* to assign a new label to the messages.



Guided Activity 12.6

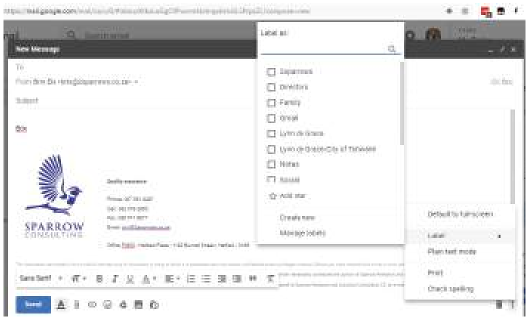
1. Select and open the email.
2. Click the X on each label you want to remove from the email.



Guided Activity 12.7

To label a message you are composing:

1. Click the *Menu* button in the lower right corner of the message.
2. Move the pointer down to *Labels*.
   A slide-out menu will show you labels you have already created, if any, and give you the option of creating a new label.



Guided Activity 12.8

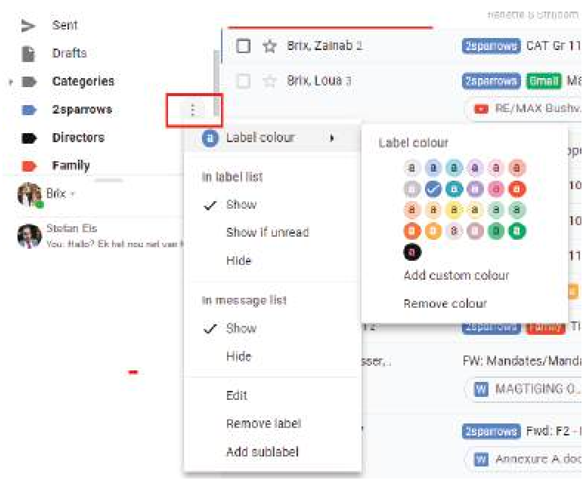
You can delete an entire label you no longer want at any time. Doing so will not delete the emails that carry that label, however. To delete a label:

1. Open or select any message or conversation in Gmail.
2. Click the *Labels* icon in the toolbar.
3. In the menu, click *Manage labels*.
4. Scroll down to the *Labels* section and under the *Actions* column, click REMOVE.
5. Click *Delete* to confirm the action. Gmail will delete the label and it will be removed from any messages it applies to.

Guided Activity 12.9

All labels are coloured by dark grey text on a light grey background. To customise your label colours, you must do the following:

1. Move your mouse over the chosen label.
2. Click on the *Menu* (three vertical dots in a grey circle). The *Label colour* drop-down box will appear.
3. Move your mouse over the *Label colour* option and select a text and colour combination by clicking on it.



PRIORITISING EMAILS

You can use the Gmail star system to prioritise your emails so that you can easily find the most important ones later.

By default, the Gmail stars are yellow. You can, however, change the colour as well as the type of the stars to suit your own needs.

Figure 12.2: *Gmail's star system*

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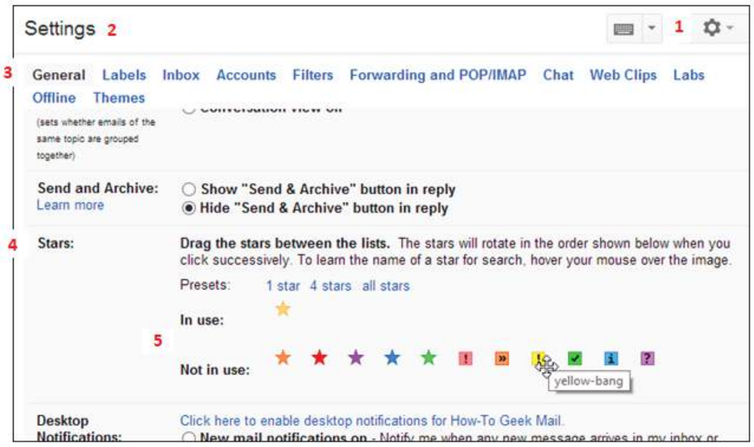
Guided Activity 12.10

To use the Gmail star system, set up your stars:

1. Click the *Settings* gear icon.
2. Choose *Settings*.
3. Click on the *General* tag.
4. Scroll down to *Stars*.



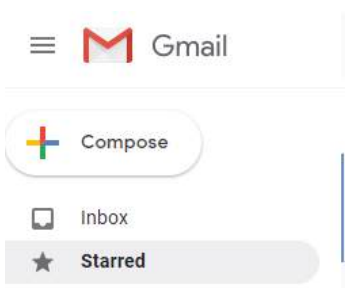
5. Drag icons from the *Not in use* section to the *In use* section to add different types of stars.

Guided Activity 12.11

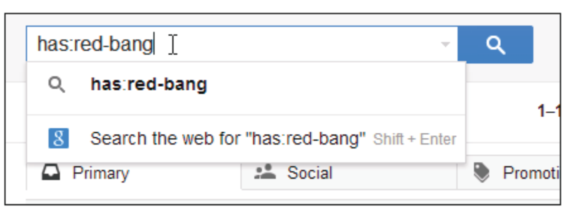
1. Add a star to a message by clicking on the star next to the email.

If you click it once, you will get the default yellow star. By cycling through the stars (clicking), you can choose which star to add. If you star a message while it is open, only the first star type is applied.

2. To find a starred message, click the Starred label on the left side of the main Gmail window.



3. To find a message with a particular type of star, search using "has:" with the star's name (e.g. "has:red-bang").



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Guided Activity 12.11

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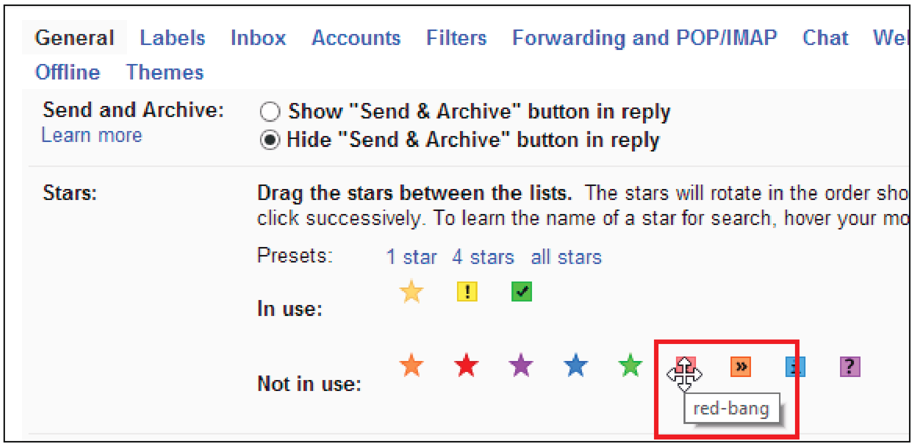
4. To find out what the name of the star is:
   a. Click the *Settings* gear icon.
   b. Choose *Settings*.
   c. Click on the *General* tag.
   d. Scroll down to *Stars*.
   e. Hover your mouse over the specific type of star.



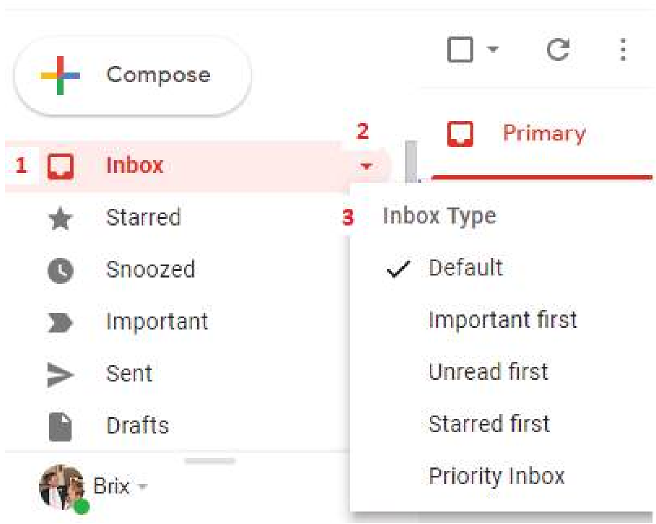
Gmail's inbox will help you to sort out your incoming emails by setting priorities. There are five sections, namely:

1. Default: You can choose which emails you want to see first.
2. Important first: Puts all email marked important at the top of your inbox.
3. Unread first: Puts all unread email at the top of your inbox.
4. Starred first: Puts all starred email at the top of your inbox.
5. Priority inbox: This option lets Gmail sort and prioritise emails for you.

Guided Activity 12.12

To decide which type of email you want to see at the top of your inbox:

1. Open your Gmail inbox.
2. Click on the down arrow on the left.
3. Choose an option.



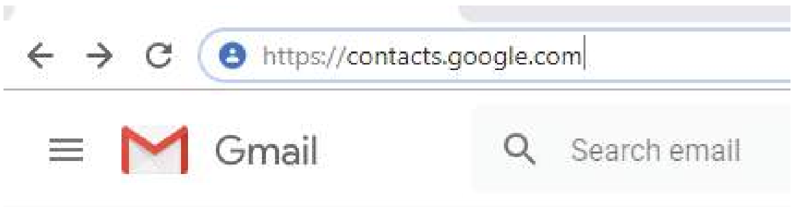
DISTRIBUTION LISTS

A contact group (formerly called a “distribution list”) consists of a list of email addresses without having to add each name to the To, Cc, or Bcc lines of an email individually. Its main purpose is to allow a user to automatically share a message to all the recipients on the distribution list at the same time.

You can, for example, have a personal distribution list called “family” that includes the email addresses of all your relatives. Rather than typing or selecting each address, you just use the list address. A distribution list is different from an email list in that members cannot directly send emails to other members who are on it.

Guided Activity 12.13

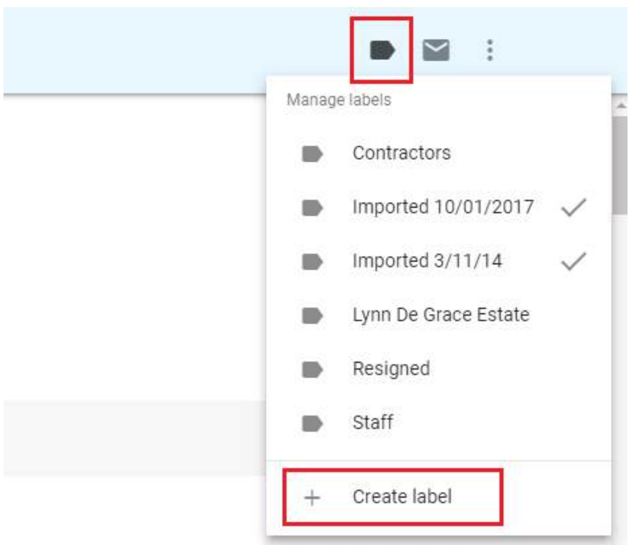
1. Open the Google contacts page by typing https://contacts.google.com/ in your web browser.



   This will open a page with all your contacts’ details on it.

2. Select your contacts by hovering your mouse over each contact’s profile picture or placeholder.

3. Click the checkbox that appears below your cursor and repeat this process for each contact you want to add.
4. Click the *Labels* icon in the upper right corner of the page. A dropdown menu will appear.



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Something to know

This can also be done by clicking on the waffle icon (⠿) and then clicking on *Contacts*. You may need to click on *More* first to see the link to your contacts.

Guided Activity 12.13

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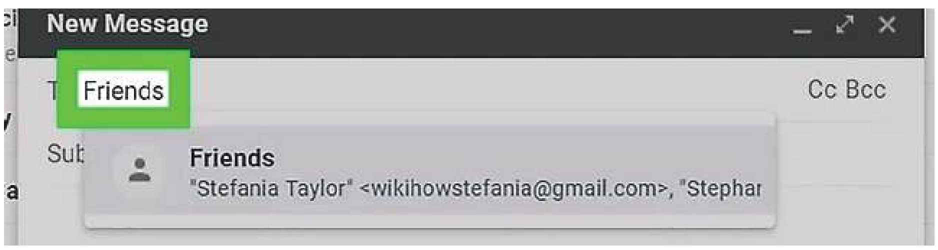
5. Click on *Create label*. This prompts a pop-up window.

6. Type in the name of the distribution list, for example “Friends”. This is the name that you will use in the *To* field when you want to send an email.
7. Click *OK* at the bottom right corner of the pop-up window. This will save your list of contacts as a label.

To send an email to your distribution list, you will follow the normal email procedures, typing in “Friends” in the *TO* field.



FILTERS

Filters help you to set up rules for how Gmail should handle your incoming emails, for example it can send an email to a label, or it can archive, delete, star or automatically forward the email.

Filters are relatively easy to set up using search criteria in Gmail itself.

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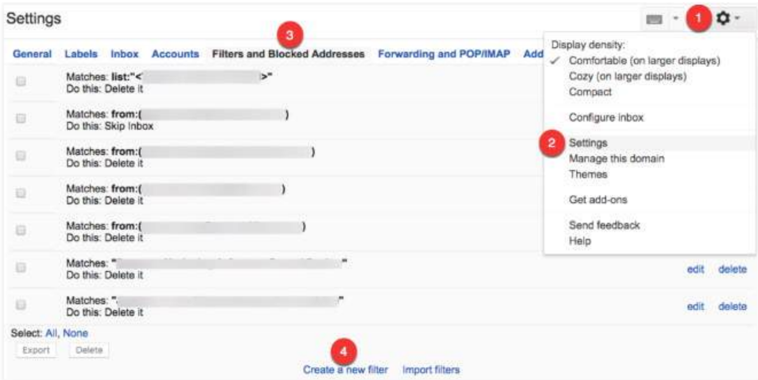
Guided Activity 12.14

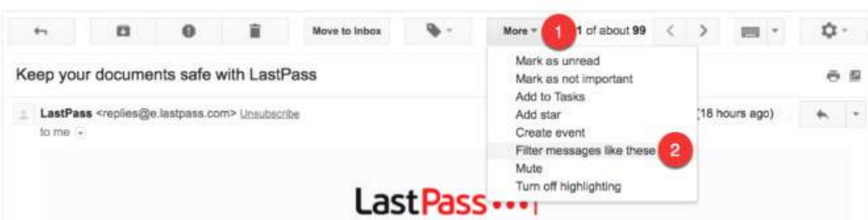
1. Click the *Settings* gear icon near the top right corner of the Gmail screen.
2. Follow the *Settings* link in the menu that comes up.



3. Select the *Filters* and *Blocked Addresses*.
4. Click the *Create a new filter* link.
5. Complete the filter creation form.

1. *From*: Filter emails sent from a specific email address.
2. *To*: Filter emails sent to a specific email address.
3. *Subject*: Filter emails that use a specific subject line.
4. *Has the words*: Filter emails that contain specified keywords.
5. *Doesn't have*: Filter emails that don't contain specified keywords.
6. *Has attachment*: Filter emails that include an attachment.
7. *Don't include chats*: Ignore Hangouts chats when applying filters.
8. *Size*: Filter emails larger or smaller than a specific size.

You can also filter specific messages by opening the email, clicking the More button and selecting Filter messages like these.



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Guided Activity 12.14

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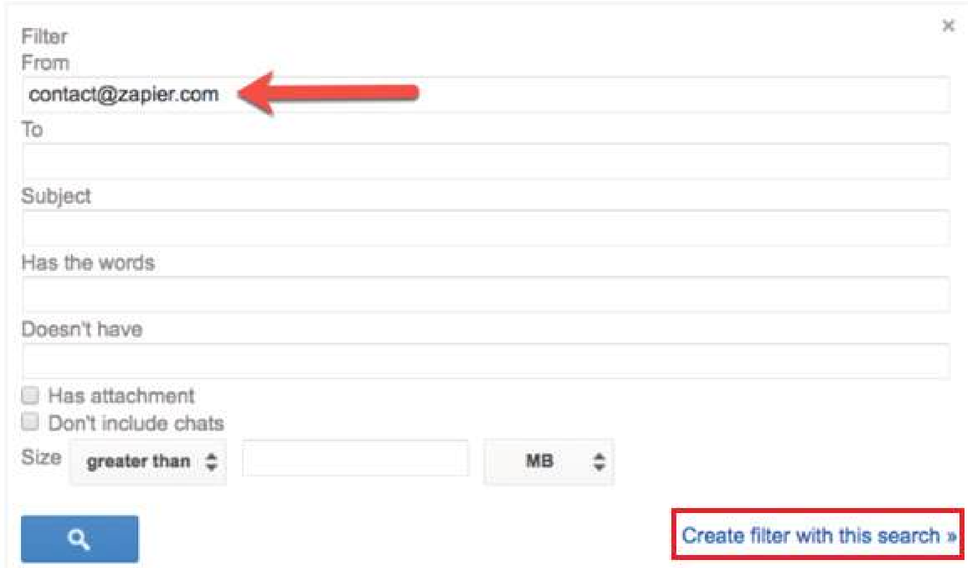
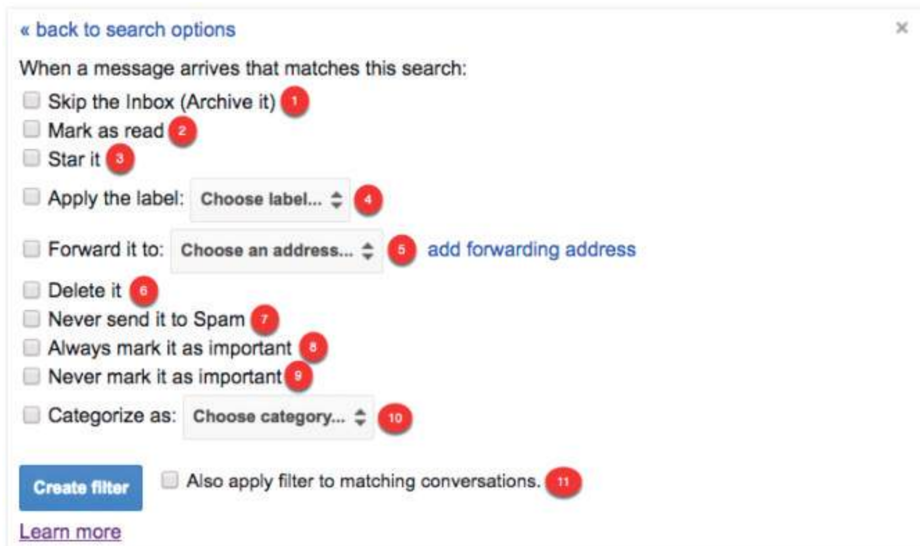
If you used this option while in a specific email, Gmail automatically fills in the "from" email address for you:

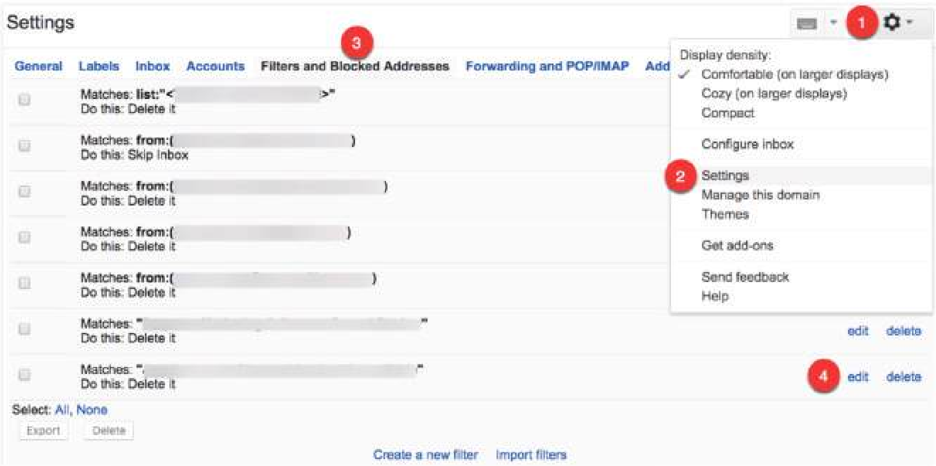


6. After selecting the criteria for your filter, click the *Create filter with this search* link to specify what Gmail must do with emails that match this filter.



1. *Skip the inbox*: Archive the email so that it does not appear in your inbox.
2. *Mark as read*: Have the email appear in your inbox as an already-read item.
3. *Star it*: Automatically star the email.
4. *Apply the label*: Apply a specific label to the email.
5. *Forward it to*: Automatically forward the email to a different email address.
6. *Delete it*: Send the email to the trash.
7. *Never send it to Spam*: Prevent Gmail from tagging the email as spam.
8. *Always mark it as important*: Automatically tag filtered emails as important.
9. *Never mark it as important*: Tell Gmail not to tag filtered emails as important.
10. *Categorise as*: Automatically categorise filtered emails.
11. *Also apply filter to matching conversations*: Automatically applies the selected conditions to every email in your account that matches the selected filter criteria.

Guided Activity 12.15



1. Click the *Settings* gear icon.
2. Choose *Settings*.
3. Select the *Filters* and *Blocked Addresses* tab.
   Find the filter you want to edit or delete.
4. Click the *Edit* link to update the filter criteria and behaviour or click the *Delete* link to delete it.

Activity 12.2

Answer the following in your own words:

1. Give three examples of email applications.
2. What are the benefits of using labels in Gmail?
3. Describe how you set up a filter in Gmail.

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

1.1 Which of the following is NOT electronic communication? (1)
- A. Instant messaging
- B. Teleconferencing
- C. Faxing
- D. Social networking

1.2 Which of the following can be done with Gmail labels? (1)
- A. Creating filters
- B. Prioritising emails
- C. Archiving emails
- D. Organising emails

1.3 What does the following process show? (1)

Step 1: In the section of the *Labels* tab, click *Edit* in the *Actions* column for the label you want to move.

Step 2: Check *Nest label under* and select a destination from the drop-down menu.

Step 3: Click *Create* or *Save*.

- A. How to move a label under another label
- B. How to filter your inbox
- C. How to move an email under a label
- D. How to star an email

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REVISION ACTIVITY *… continued*

1.4 When would you use the Gmail star system? (1)
- **A.** To highlight all your important emails
- **B.** To delete an email
- **C.** To organise all your emails
- **D.** To find a specific email

1.5 You and a specific group of your friends share inspirational quotes with each other every day by email. Which of the following make this process easier? (1)
- **A.** Star system
- **B.** Labels
- **C.** Distribution list
- **D.** Filters

QUESTION 2: TRUE OR FALSE

Choose the answer and write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

- **a.** When it comes to a password, try to choose a password that is <u>short and easy to remember</u>. (1)
- **b.** Artificial intelligence is a type of <u>electronic communication</u>. (1)
- **c.** To keep your system simple and easy <u>add all the labels and categories you need</u>. (1)
- **d.** You can remove a label from an email by clicking on the <u>"Remove" option</u> on the label you do not want. (1)
- **e.** All labels are coloured by <u>a dark grey text</u> on a light grey background. (1)

QUESTION 3: MATCHING ITEMS

Choose a term/concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1A). (5)

COLUMN A	COLUMN B
3.1 Gert wants all his unstarred emails at the bottom of his inbox.	**A.** Sent inbox **B.** Default
3.2 Bianca uses this to manage the prioritisation of her emails.	**C.** Important first **D.** Inbox
3.3 When Malusi opens his email he wants specific emails to appear first.	**E.** Unread first
3.4 Priyanka wants to make sure that all her new emails are at the top of her inbox.	**F.** Starred second **G.** Priority inbox
3.5 Gina wants to put all her important email at the top of her inbox.	**H.** Starred first

QUESTION 4: FILL IN THE MISSING WORDS

Fill in the missing word(s) in the following statements. Provide only one word for each space.

- **a.** A distribution list consists of a list of email addresses in a standard email program or on a **i)** ____________ system. Its main purpose is to allow a user to automatically **ii)** ____________ a message to all the recipients on the distribution list at the same time.
- **b.** When someone sends you an email, you need to **iii)** ____________ to the email, since the **iv)** ____________ will not automatically know if you received the email or not. (4)

… continued

REVISION ACTIVITY

… continued

QUESTION 5: SCENARIO-BASED QUESTIONS

Nandini is a website designer. She has not maintained her email account for a long time. But recently a client of hers has requested that she forward them an email that shows confirmation for a bank account payment. With more than 600 emails to sift through, to find the exact email her client is looking for will take time.

5.1 Mention one thing that Nandini can do to better manage her emails. (1)

5.2 Nandini wants to organise and separate all her emails under the following labels:

- Clients
- Personal
- Family
- Friends

a. What process must Nandini follow to create a filter that will mark all her clients' emails as important? (6)

b. What process must Nandini follow if she wants to create a label for all her clients' emails? (6)

c. What process must Nandini follow if she wants to create a distribution list for her following clients? (8)

- mpumi4sure@gmail.com
- Jared.masters@creativedesigns.co.za

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU …	YES	NO
1.	Describe how to manage email inboxes in relation to organising by using folders or labels?		
2.	Prioritise emails in your inbox?		
3.	Create a distribution list?		
4.	Register a web-based email address?		

TERMS 2, 3, 4

CHAPTER 13 INFORMATION MANAGEMENT

By the end of this chapter, you will be able to:

- Create a task definition
- Gather data and information using various tools
- Describe how to evaluate data to ensure that it is reliable and accurate
- Discuss how to evaluate the validity of a website as a source of information
- Evaluate information for accuracy and validity
- Describe the role of spreadsheets and databases in data handling
- Discuss tools and techniques for data handling and how to extract data
- Describe how to analyse trends and patterns in data

INTRODUCTION

Data is all around us and we create data with our every action, both online and in the real world. Data can be used in various industries to draw many insights into the way people shop, travel and interact with companies, brands and other people.

The process of collecting, processing and presenting data and information is called information management and consists of three main steps:

1. Input: This is the first step of the information management process, consisting of the identification of the main problems and collecting relevant data.
2. Process and analyse: Once all the data has been gathered, it is converted to information by processing and analysing the data.
3. Output: This is the final step in the process and consists of presenting the information in such a way that it can be easily understood by other people.

Solving problems successfully or completing work on time largely depends on your ability to find, process and present the correct information. Before you start looking for information, though, you will need to work through your task to make sure that you understand it fully and that you know what information you need and where to find it. You can do this by following the steps in the information cycle shown in Figure 13.1.

Figure 13.1: *The information cycle steps*

You need to keep in mind that the order of this cycle is not set in stone and you may need to go back to refine or redo previous steps, but the further along in the cycle you get, the harder it is to go back and do earlier steps. This is why it is important that you do steps one and two as well as possible before moving on.

If you do not have the right information, you will not be able to solve your problem or complete your task properly, be it completing your PAT or even writing this textbook.

In this section, we are going to focus on the following steps of the cycle:

1. Defining the task
2. Formulating the questions
3. Identifying and evaluating questions, information and information sources (such as websites)
4. Analysing and processing the data

UNIT

13.1 Defining the task

It may seem obvious but before you can successfully complete any task or investigation, you need to understand exactly what it is that you are going to do. This is why you need to carefully consider the task you have been given by breaking it down into smaller sections and describing it in your own words, called the task definition.

To understand what is expected of you, you must do the following:

1. Read the instructions. Does the task call for a report? How will you gather the information?
2. Rewrite the question in your own words. Get to the core of the question. Rewriting helps you understand exactly what you need to do so that you do not waste time looking for the wrong data. It also helps you decide what type of data you need and how to gather it.
3. Identify the key terms. Look for verbs or subject-specific terms in the task (verbs like "report back" give guidelines on how you should present information).

DEFINING THE TASK

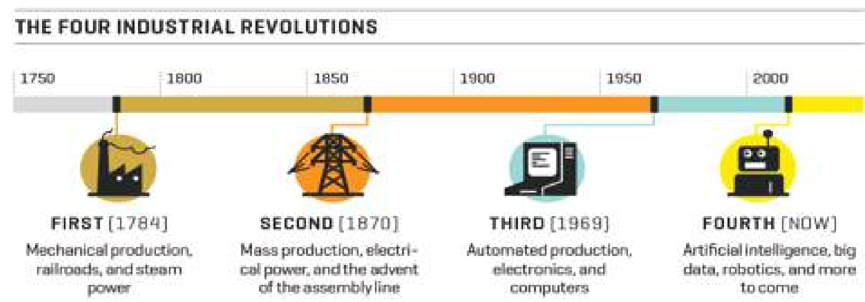
Look at the image below, taken from the 2018 PAT.

TECHNOLOGY AND SKILLS FOR A CHANGING WORLD

"While our current world is shaped fundamentally by math and science, which we learn in school, our future world will be a digital world, where our lives aTinkaped by computers and connected devices." ~ Tink Tank

The *focus question* that you are required to answer is:

How does the Fourth Industrial Revolution (4IR) impact our daily lives and what would the roles of technologies and trends such as the Internet of Things (IoT), Augmented Reality (AR) and Artificial Intelligence (AI) be?



Source: http://fortune.com/2016/03/08/davos-new-industrial-revolution/

Your task is to:

- Investigate the impact of the 4IR in terms of any TWO of the following: IoT, AR and AI. Questions that you may want to answer:
  - What do each of the two terms/concepts entail?
  - What technology is involved?
  - How does each of the two work?
  - For each of the two, choose and discuss one place where it is implemented in detail, e.g. education
  - Where is it used? For each, choose and discuss one area in detail, e.g. education, manufacturing
  - What skills will be required?
  - Advantages/disadvantages
- Gather and analyse data relevant to the investigation
- Identify a suitable audience (such as the learners in your class) and present your research and findings using a report that would be suitable for use by the specific audience

Looking at the task outline, you can begin to define what you need to do. In this case, you would need to choose two technologies from the Fourth Industrial Revolution (4IR) and investigate their impact on modern life.

The task outlines various questions you can ask yourself to further refine your search for information. You are then instructed to gather the data and analyse it and, finally, to present your findings in a report. With this information, you will be able to start your research.

Something to know

Your PAT task description should be about 200 words.

To understand how important it is to make sure that you know what you are being asked to do, read through the scenario below.

UNDERSTANDING TASKS

Julia, Mbali and Shiven are on the organising committee for the Grade 12 dance. They have each been given a task to complete to better help the organisers understand what the Grade 12s want:

- Julia has been asked to find out how many Grade 12s would be interested in bringing a partner from outside the school to the dance and to give a report to the teacher supervisor, Mr Smith.
- Mbali has been asked to find out what theme the Grade 12s want for the dance so that she can let the decorating committee know what they need to buy.
- Shiven has been asked to find out about the music.

Julia begins by breaking down her task. She decides that the best way to go about gathering the information is through an online survey. She first has to design the survey and then distribute it. She decides to use Google Forms for her survey and sends the link to the Grade 12 WhatsApp group. She asks people to please fill in her survey by Friday. Once she has this information, she downloads the report that Google Forms generates and gives it to Mr Smith.

Mbali also decides to break down her task but immediately runs into a problem. She has no clear idea of what the decorating committee is willing to spend on decor and they have not given her suggestions for themes. Mbali realises that if she were to ask the Grade 12s what theme they would want, she would end up with a different response for each person.

Mbali will need to get more information before she can complete her task. She decides to ask the decorating committee the following questions:

- Are there any theme suggestions?
- Is there a budget and how does that limit the theme?

The decorating committee tells her that they have three theme ideas. Now that she has this information, she can define her task. She decides a simple survey will also help her get the information the decorating committee needs.

Shiven's task is too vague. He will need to get more information on the following:

- Is the school willing to pay for a band or DJ? Or should he ask for a learner volunteer?
- Will there be sound equipment available or will the performers need to provide their own?
- What kind of music do the Grade 12s want at the dance?
- What kind of music will the school not allow?

Only when he has this information, will he be able to define his task and give the appropriate solution.

Activity 13.1

Define a task definition and a focus question for the 2018PAT.

13.2 Finding information

INFORMATION-GATHERING TOOLS

After you have identified the task definition, you will have to gather data and information. In this phase, you must consult different types of information sources to understand the subject of your investigation better.

Information can be gathered using the following tools:

- Electronically: internet
- Printed media
- Surveys and questionnaires

Activity 13.2

Using the task definition you created in Activity 13.1, answer the following questions to determine how you will research the topic and obtain the information you need for the task you defined.

1. Which type of information gathering tools will you use?
2. Identify at least two websites on the subject that you can use to compare your findings against.
3. Develop a questionnaire with 10 questions that you can use to obtain school-specific information.
4. How will you assess that the information you get is accurate, sufficient, valid, current and objective?
5. Discuss your work with another group and compare results.

UNIT

13.3 Quality control of information

A large majority of the content on the World Wide Web is user-generated, meaning it is created by people of any age, educational background and experience. This means that when you are using the WWW to research your PAT, you will need to carefully evaluate the information you find. In this chapter, you will learn about how to evaluate information to make sure that it is reliable and accurate.

EVALUATE QUESTIONS

When you are looking for the data you need to complete your task, you can ask yourself questions to guide you. There are two main types of questions, namely open-ended and closed-ended questions.

Closed-ended questions have a limited set of possible answers, such as “yes”, “no” or “A”, while open-ended questions need more complex answers. You should not make the mistake of thinking that closed-ended questions are simple. Some closed-ended questions, for example “Are amoebas single-celled organisms?” have a specific answer, but not everyone can answer them.

While both types of questions are good for gathering data, closed-ended questions are only useful in the early stages of the data gathering process when you are trying to find as much data related to your topic as possible. This is because they put an end to the conversation. Closed-ended questions are good for surveys or where your respondents do not have the time to spend thinking about their answers.

When using closed-ended questions for research, ask yourself: “Does this data match my topic?” If it does, make a note of it, if it does not, discard it. You will then use open-ended questions to refine the data further and make sure that what you have found helps you complete your task.

In order to refine your data using open-ended questions, you need to decide what type of open-ended questions you will ask. There are several types of open-ended questions, for example:

- Questions that can be answered explicitly by facts, for example, questions starting with words such as “what”, “when”, “where”, “who”, or “how many”.
- Questions that will help you to examine, explore and query, such as questions starting with “why” and “how”.
- Questions that will help you to adjust, change or predict, such as questions starting with “if”, or “what if”.
- Questions that will help you to make a judgment, critique, review or find meaning of some sort, such as questions starting with “would it be better if”, “what recommendation”, “how can I determine”, or “what would be the best way”.
- Open-ended questions allow you to get more from the data than you anticipated when you use these questions to interrogate that data.

You can also use a combination of open-ended and closed-ended questions in questionnaires. You will most likely be asked to create a questionnaire for your PAT, so knowing how to ask these questions when you are researching will help you to set them when the time comes.

Tips for asking questions:

- Always start an open question with "how" or with questions beginning with "w": "what", "where", "when", "which" or "who".
- Try to avoid "why" questions, because human nature leads people to make up a rational reason even when they don't have one. If you want to get more information, rather say "please tell me more about that".
- Do not start with "was" or forms of the verbs "to be" and "to do".
- Try to get the person you are asking a question to tell a story, rather than giving one- or two-word answers.
- If you cannot avoid asking a closed-ended question, follow it up with an open-ended question such as "what else can you tell me about it".
- You can get an open-ended answer from a multiple-choice question if you add "other" as an option.

EVALUATE INFORMATION SOURCES

Once you have defined your task, you can begin to gather your data and put it together to turn it into information. To do this you will need to know what types of questions to ask to get the information you need, as well as evaluate the quality of the information and the websites where you find it.

EVALUATE WEBSITES

When you are doing research online, it is important to think about the websites you are using. Not all websites are created equal and many may contain misleading, or even false, information. Much as you evaluate the quality of the information you receive, you must also evaluate the quality of the websites you visit and use.

There are some key questions you can ask yourself about the websites you are visiting and using, for example:

- Who supports this website?
- Who is this website aimed at?
- Who is the author? What are their credentials?
- How is the content organised? Do all the links on the pages work?
- How easy is it to navigate around the website? Is the design pleasing and attractive? How quickly does the site load?
- What is the grammar and spelling like?
- Is the content objective or does it support a single opinion?

Something to know

Low-quality websites might not only give you bad information, they could also be a way for malicious software such as viruses, spyware or malware to get onto your computer.

These are all important questions to ask when looking at a website. Let us look at each point in detail to understand why you should ask these questions.

- Authority: Look at the web address of the website. The domain name can help you determine if the information has been published by a credible source, for example .ac says that it is published by a university, .gov refers to the government, etc.
- Look for the author's name. If they choose to remain anonymous, ask yourself why. A quick Google search will also show you if they have any qualifications in the field they are writing about. This is especially important in scientific or technical articles, since you would want an expert telling you how to do something or explaining something to you.

- The grammar and spelling on the site are also extremely important. If a website is filled with grammar and spelling errors, it may not be the most authoritative or authentic source for the information you are looking for.
- Affiliation and objectivity: You should always be aware of who owns or financially supports it. This could be a good indicator of whether there will be bias in the information. Take a close look at the content and try to work out if it is **biased** or supports one opinion too heavily. Biased content is the opposite of high-quality content, as it has a very narrow view on a subject.
- Try to use websites supported by, or linked to, established institutions such as government agencies, non-profit organisations or educational institutions. Remember, objective sites present information with a minimum of bias.
- Audience: You should also make sure that the website's target audience (the people it is aiming its content at) is appropriate for your needs. A website aimed at young children will not help a university student with their research, while a website aimed at giving information to doctors will not help an engineer in their research. Make sure that the content you are looking at is appropriate for your needs.
- Content: Another important thing to look at is how the content of the website is arranged. Does it make sense or is the information scattered all over the place? Is it easy to find what you are looking for or do you have to spend ages trying to figure out the website's navigation? Also check to see if all the links work, in articles and on the website itself. Broken links are often a sign of a badly maintained or spammy website.
- Currency and design: Is the website designed well and does it look professional? If not, this could be a sign that the website is old and outdated or just badly designed, which could mean that the information you find there will be the same. Apart from being very frustrating, websites that load slowly are often badly designed. This could be an indicator of badly written or inaccurate information.

EVALUATE INFORMATION

It is most likely that your primary source of information will be the internet. Understanding the quality of information is important, since the internet is a place where anyone can upload information or make claims. You will need to look carefully at the information you use to discuss the task. You should look at the source of the information and try to find out who has an interest in this data (this is especially important when it comes to scientific papers). You should also think about what people will gain by spreading certain types of information on the internet.

There are five key things that determine the quality of information:

- **Authority:** This is based on who created the information. If the video you are watching was not created by someone with knowledge in the field, then the information they are giving you might not be the best out there. Always do research on the people who give you information. Authority also indicates that the information is accurate.
- **Accuracy:** Look for how complete the information is and compare the information from one source against the information from a few others. If the facts match, you can be fairly sure that the information is accurate. Make sure that there is something backing up the facts you receive. Check the source of those facts as well.
- **Currency:** Check how current the information is. More up-to-date information tends to be more accurate. A technology blog from 2010 will be less relevant, and therefore less accurate, to your research than one from 2017. Information from the last ten years is seen as current.

- **Objectivity:** Make sure that the author is being objective. Look out for information that is sponsored by a company (that is, if it is a type of long-form advertisement called a sponsored post). This information may be biased towards the facts that the sponsor wants to represent to make their product or information seem like the best.
- **Relevance:** Lastly, look at how well the information covers the topic you are researching. If the information only covers a small portion of the topic, then it is possibly not the best information to use. You need to look for information that answers about 80% of any question you are trying to find the answer for.

Once you understand how to evaluate the quality of information, you can be confident that you will present your information accurately and reliably.

Activity 13.3

Answer the following questions in your own words.

1. What are the key questions to ask when evaluating a website?
2. Why must you evaluate information?
3. What are the five key things that determine the quality of information?
4. Determine if the following questions are open-ended or closed-ended. Change the closed-ended questions into open-ended questions.
   a. Is it hot outside?
   b. Have you already completed your homework?
   c. How does this make you feel?
   d. Do you have plans for the December holidays?
   e. What are your plans for when you finish school?
   f. Do you like that person?
   g. How do you book tickets for a flight?
   h. Is that your final answer?
   i. Are you satisfied with the results?
   j. Did you know that…?

13.4 Spreadsheets and databases in data handling

Spreadsheets are documents that have numbered rows and lettered columns that structure data. Examples of spreadsheet software include Microsoft Excel, Google Sheets, LibreOffice Calc and iWork Numbers. You can use spreadsheets to manipulate and organise raw data before putting it in a database, as well as to:

- Do mathematical calculations (adding totals, working out averages and percentages, etc.)
- Display data using charts and graphs
- Define and refine data by breaking it down or splitting it (for example, taking a single row and column of data and splitting it into many columns using a function)

Spreadsheets can also be used as a modelling tool, that is, you can use them to predict **trends**. As an example, if a company wants to find out what would happen if it reduced the price of one of its products, it could use a spreadsheet to analyse the changes to its profit margins.

Databases, on the other hand, can do exactly what a spreadsheet can, but with more data, since spreadsheets are limited in the amount of data they can hold. Databases are also usually kept on a server, so they can use the bigger processing power of the server to do large and complex procedures and calculations with the data faster than a spreadsheet can.

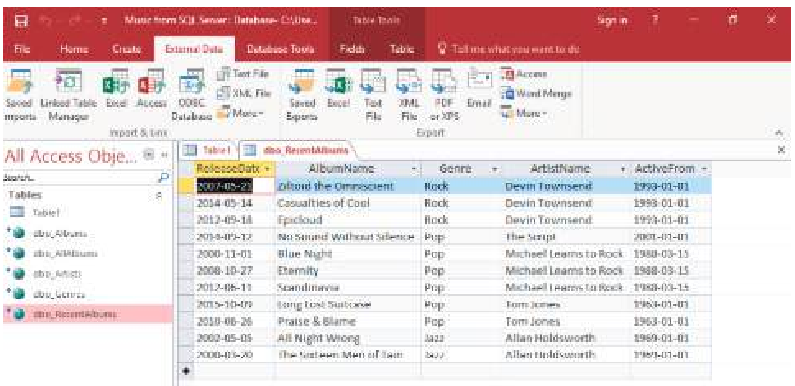


Figure 13.2: *MS Access database*

Databases can also better analyse and connect data than spreadsheets, which makes it easier for users to search for, and extract information from, the data. Databases are also designed to connect data more easily across multiple tables.

However, other than the amount of data and the speed of processing, databases and spreadsheets are very similar. In fact, most reports are generated in a database and then exported to a spreadsheet to be displayed. Some website databases need data to be uploaded in spreadsheets saved in a special format called comma separated values (*.csv).

Activity 13.4

Answer the following questions in your own words.

1. What is the main difference between a database and a spreadsheet?
2. When will you use a database rather than a spreadsheet?
3. What is data handling?
4. Give three examples of what you can use spreadsheets for.
5. What can databases do better than spreadsheets?

UNIT

13.5 Processing data

You need to process data or manipulate raw data to draw information from it, that is, you need to **crawl** through and analyse the data to draw the most relevant/appropriate information from the data.

There are several techniques or tools you can use to do this. These will be discussed below.

TOOLS AND TECHNIQUES

Manipulating data is the process of **sorting**, arranging or moving the data without changing it. To manipulate data, you should first keep the following in mind:

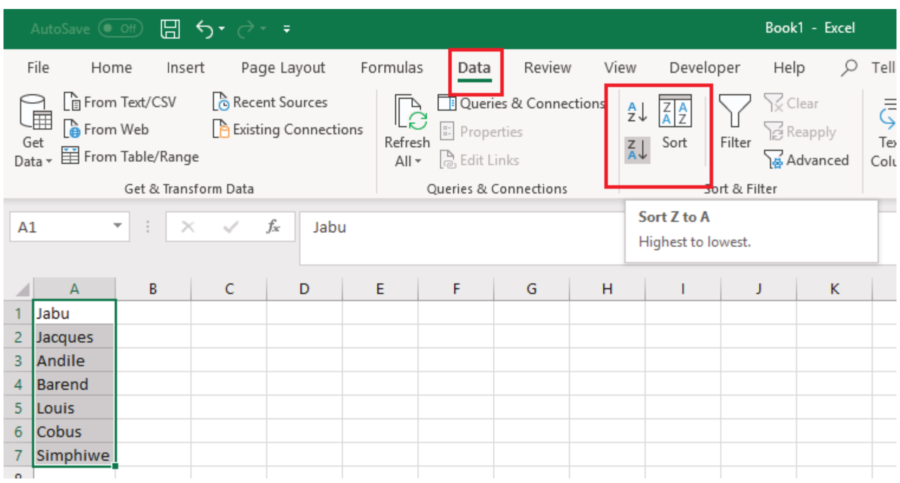
- You need to know the features of the programs you are using and how to use them.
- Make sure that the data is in the correct format for what you need to do. If it is not, you will need to reformat it or export it into the most appropriate format.
- You might have to use more than one tool to process the data correctly.
- The tools you can use include sorting, formulae and functions and queries. Sorting changes the order of the data. It can be done to numerical data (sorting numbers from lowest to highest) or strings (sorting text alphabetically). This is mostly done in spreadsheets.

Guided Activity 13.1

This is an important function to use in Excel as you might want to arrange data alphabetically, from highest to lowest, or order it by colour or icon. This will help you to visualise and understand your data better.

You can, for example, sort text from A to Z or Z to A, numbers from smallest to largest or vice versa, and dates from oldest to newest, or newest to oldest. You can also sort according to your own custom list or format.

1. To sort text:
   a. Select a cell in the column you want to sort.
   b. On the *Data* tab, in the *Sort & Filter* group, do one of the following:



- To quick sort in **ascending order**, click [A→Z] (Sort *A* to *Z*).
- To quick sort in **descending order**, click [Z→A] (Sort *Z* to *A*).

… continued

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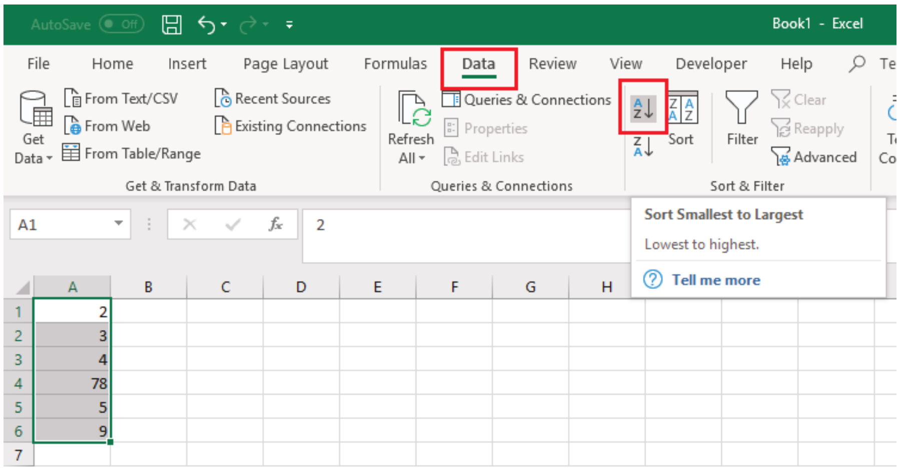
2. To sort numbers:
   a. Follow steps one and two above.



- To sort from low to high, click (Sort *Smallest* to *Largest*).
- To sort from high to low, click (Sort *Largest* to *Smallest*).

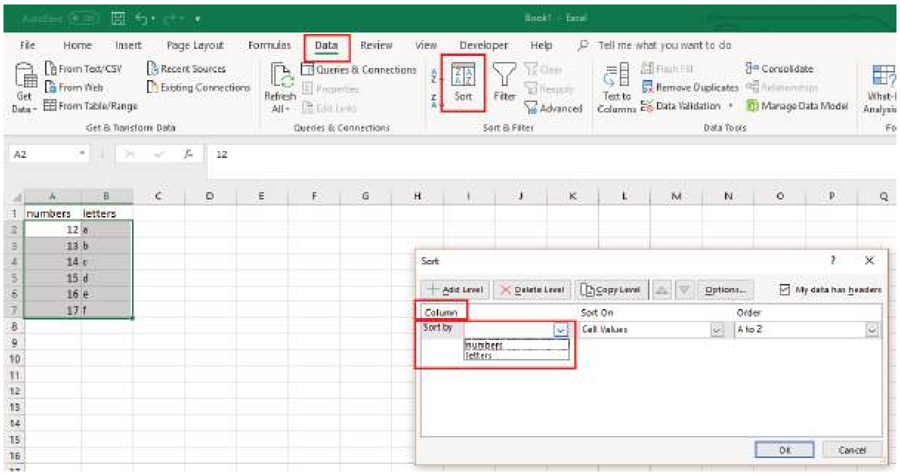
3. To sort dates and time:
   a. Follow the same procedures as above.
4. To sort more than one row or column:
   a. Select any cell in the data range.
   Note: For the best results, the range of cells that you sort should have column headings.

	A	B
1	Name	Department
2	Pruitt, Barbara	Sales
3	Fife, Grant	Marketing
4	Snook, Anthony	Operations
5	Horn, Frances	Finance
6	Barrett, Alicia	Credit
7	Larson, Lynn	Payroll
8	Brown, Charity	Sales
9	Bartels, Freddy	Marketing
10	Harding, Marcella	Operations
11	Reuter, Randall	Finance
12	McDaniel, Evelyn	Credit
13	Totten, Carlos	Payroll
14	Hull, Felecia	Sales
15	Hume, Merle	Marketing
16	Bradley, Kristi	Operations
17	Carbajal, Scotty	Finance
18	Foster, Heather	Credit
19	Harris, Corinne	Payroll
20	Erickson, Tara	Sales
21	Huntington, Jon	Marketing

   b. On the *Data* tab, in the *Sort & Filter* group, click *Sort*.

… continued

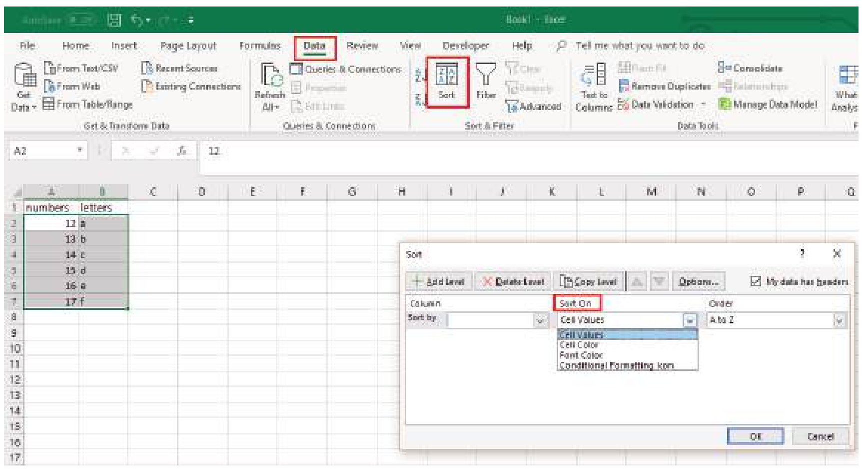
c. In the *Sort* dialog box, under *Column*, in the *Sort by* box, select the first column that you want to sort.



d. Under *Sort On*, select the type of sort. Do one of the following:

i. To sort by text, number, or date and time, select *Values*.

ii. To sort by format, select *Cell Color*, *Font Color*, or *Cell Icon*.



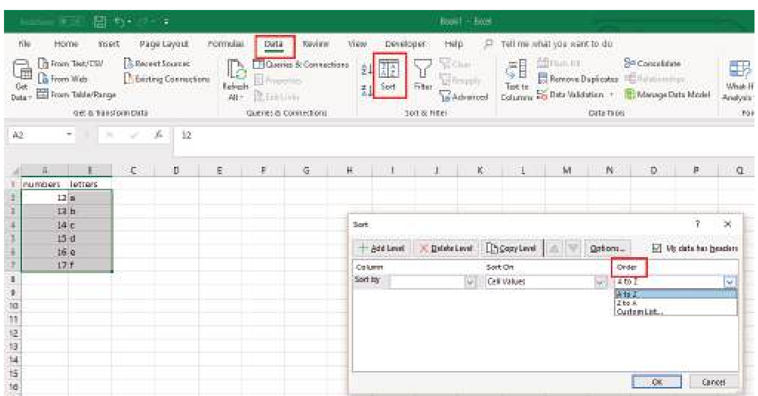
e. Under *Order*, select how you want to sort. Do one of the following:

i. For text values, select *A to Z* or *Z to A*.

ii. For number values, select *Smallest to Largest* or *Largest to Smallest*.

iii. For date or time values, select *Oldest to Newest* or *Newest to Oldest*.

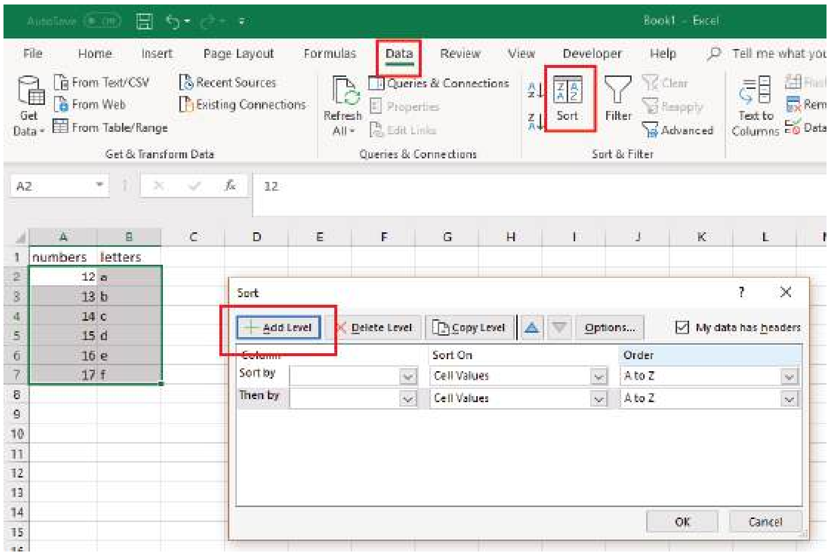
iv. To sort based on a custom list, select *Custom List*.



… continued

Guided Activity 13.1 … *continued*

f. To add another column to sort by, click *Add Level*, then repeat steps c. to e.



You can also use the various formulae and functions in a spreadsheet program (such as Microsoft Excel) to help you process data. You will have seen most of the functions in action in your practical lessons on Excel; but a list of functions is included below:

- SUM calculates the total of a range of numbers.
- AVERAGE gives the average of a range of values.
- MIN returns the minimum value in a list of values.
- MAX returns the maximum value in a list of values.
- COUNT counts the numbers in a list of values.
- COUNTIF is used for counting cells within a specified range that meet a certain condition.

Something to know

Remember, all formulas begin with an = sign in Excel.

These functions let you interpret large sets of data but are not very good for visualising the data. You can use charts and graphs to visualise data and analyse trends.

EXTRACTING APPROPRIATE INFORMATION

Within spreadsheets and **databases**, there are various tools and commands you can use to extract information, for example, for a big data set, you can use VLOOKUP in Excel.

To see how extracting information is different using Microsoft Excel and Microsoft Access, look at the case study below.

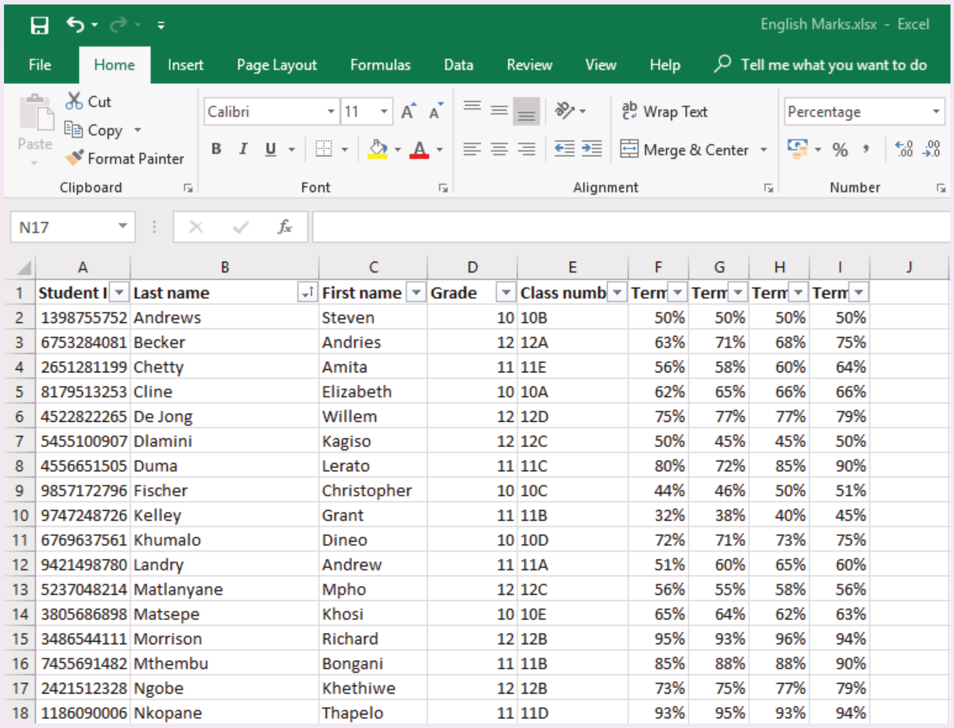
Case Study Extracting information

Mrs Dlamini is the administrative assistant at a school. The head of department for English has asked her to find the yearly average English mark for all Grade 11 learners for the previous year. The spreadsheet Mrs Dlamini is given contains the English marks for every assessment and assignment for every learner in Grade 10, Grade 11 and Grade 12.

… continued

So how would Mrs Dlamini find the information?

First, she will need to look at the data she has. When she does this, she sees that the columns have a learner ID number, their full names, grade, class number and terms one, two, three and four marks (as you can see in Figure 13.3).



	A	B	C	D	E	F	G	H	I	J
1	Student I	Last name	First name	Grade	Class numb	Term	Term	Term	Term	
2	1398755752	Andrews	Steven	10	10B	50%	50%	50%	50%	
3	6753284081	Becker	Andries	12	12A	63%	71%	68%	75%	
4	2651281199	Chetty	Amita	11	11E	56%	58%	60%	64%	
5	8179513253	Cline	Elizabeth	10	10A	62%	65%	66%	66%	
6	4522822265	De Jong	Willem	12	12D	75%	77%	77%	79%	
7	5455100907	Dlamini	Kagiso	12	12C	50%	45%	45%	50%	
8	4556651505	Duma	Lerato	11	11C	80%	72%	85%	90%	
9	9857172796	Fischer	Christopher	10	10C	44%	46%	50%	51%	
10	9747248726	Kelley	Grant	11	11B	32%	38%	40%	45%	
11	6769637561	Khumalo	Dineo	10	10D	72%	71%	73%	75%	
12	9421498780	Landry	Andrew	11	11A	51%	60%	65%	60%	
13	5237048214	Matlanyane	Mpho	12	12C	56%	55%	58%	56%	
14	3805686898	Matsepe	Khosi	10	10E	65%	64%	62%	63%	
15	3486544111	Morrison	Richard	12	12B	95%	93%	96%	94%	
16	7455691482	Mthembu	Bongani	11	11B	85%	88%	88%	90%	
17	2421512328	Ngobe	Khethiwe	12	12B	73%	75%	77%	79%	
18	1186090006	Nkopane	Thapelo	11	11D	93%	95%	93%	94%	

Figure 13.3: *English marks for senior students*

Because this spreadsheet is sorted alphabetically, it is hard to tell which learner is in which grade. In Excel, Mrs Dlamini would need to use the filters on the columns to sort the learners by grade (column D) and then only display the Grade 11 learners. She would then have to use the *Average* function to work out each student's average mark for the year in column J. At the bottom of the data in column J, she would then need to use the *Average* function again to work out the yearly average for Grade 11.

If this spreadsheet was in a database, with the exact same headings and information, Mrs Dlamini would only need to use a single **query** to do exactly what she did on the spreadsheet.

13.6 Analysing data and information

After you have collected your data and formatted it, you will need to analyse your results and compare them to a question. But how do you do this? The best way to support or disprove your question is to look at trends and patterns. This section looks at how to verify and validate your data, how to add data questions to your data to help analyse it and how to analyse for trends and patterns.

ADDING DATA QUESTIONS

Once you have verified your data, you can begin to apply the data questions to it so that you can categorise and organise the data. As you learned in Grade 10, the data questions are:

- How many?
- What is most popular?
- What is least common?
- How many more than?
- What is the average?

TRENDS AND PATTERNS

The final step in data analysis is to look for trends and patterns. While these may seem like the same thing, they are not. Trends are the general tendency of data to move in one direction over time. In a trend, the data points may vary slightly, but they all still move in the same direction.

Think of global temperatures. We all know that the temperature of the Earth is getting higher each year, in fact, scientists have been recording and comparing surface temperatures on Earth for the last 100 years. Looking at this data, we can see that while there have been some very hot and very cold days, the overall trend for global temperatures is up. You can see this in the graph in Figure 13.4.

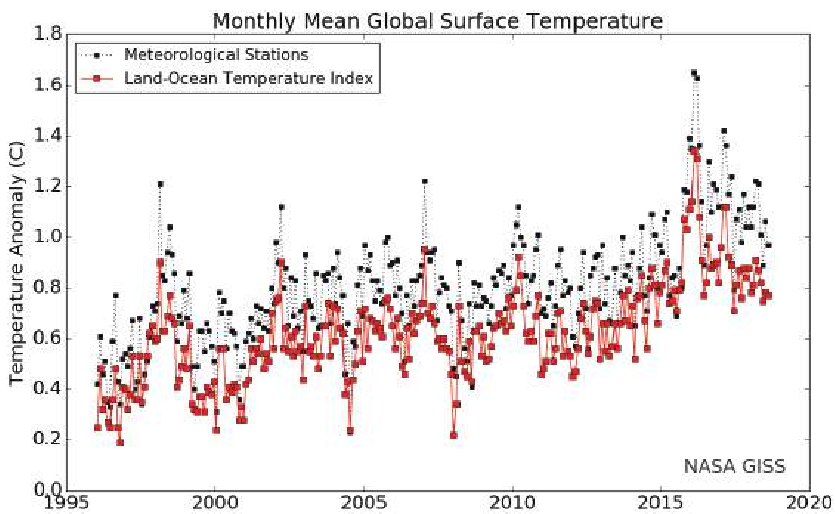


Figure 13.4: *Global surface temperature graph*

This type of trend is known as an upward trend.

Another type of trend is known as **downward trend** or downtrend. This happens when the data points move down, for example, the cost of computers and computing technology has decreased since the start of the computer age. If you look at the cost of the first home computers in the 1980s and compare that cost over time to now, you will see that the prices have come down.

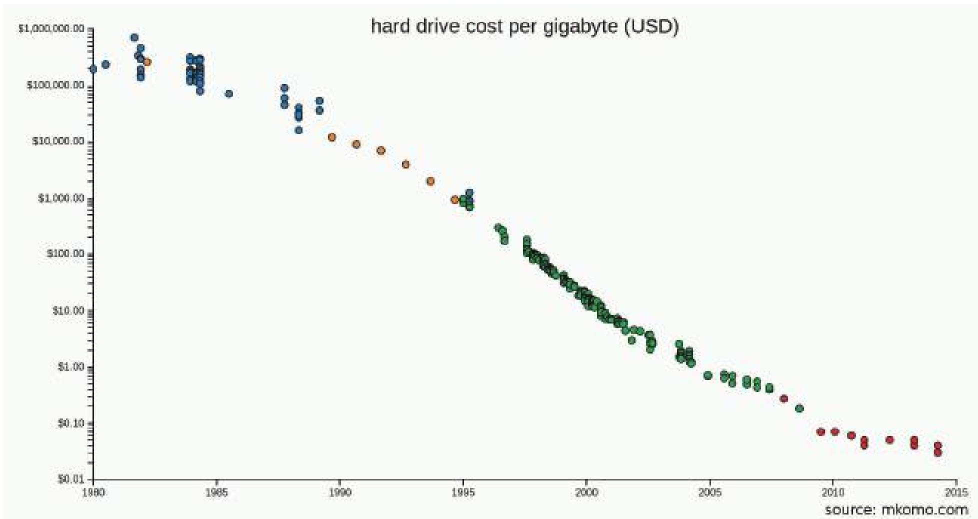


Figure 13.5: *The cost per gigabyte of hard drives in the United States of America*

You may also see a link between the cost of computing devices going down while their processing power and speed goes up as the technology improves over time. This is called a **correlation trend**.

Unlike upward or downward trends, which happen slowly over time, step changes show a massive increase or decrease in the trend. Think of the price of petrol, which can increase or decrease slowly or suddenly jump up or down depending on how much oil costs.

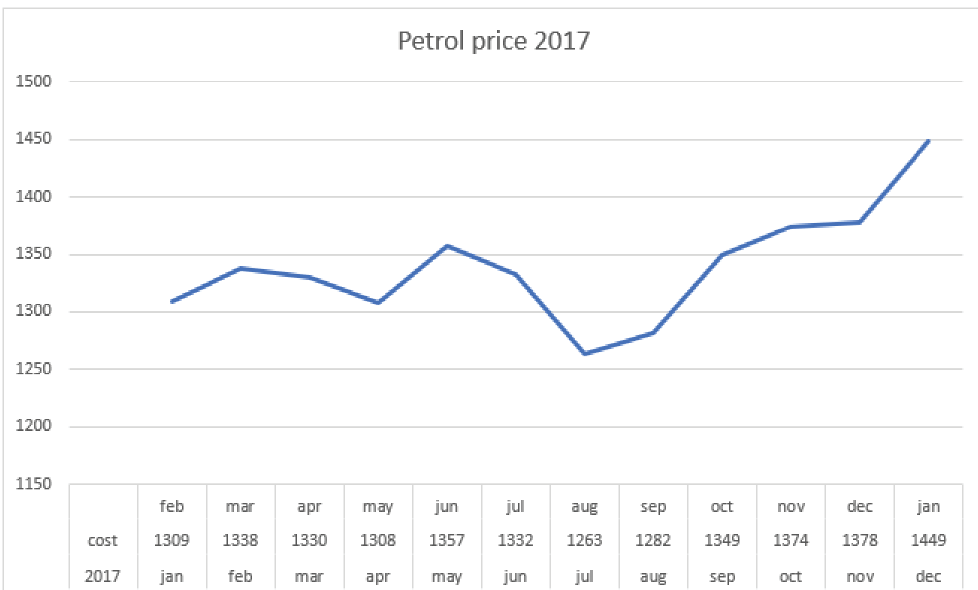
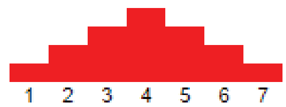
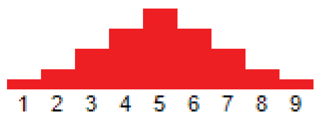
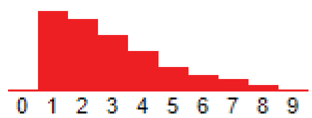
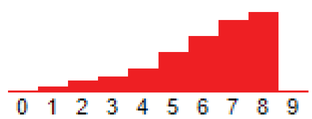


Figure 13.6: *Step chart showing the Inland price of 93 ULP in 2017*

Other trends include daily and seasonal cycles.

Patterns are a little different. Data does not always move up or down over time. Data can also repeat itself in a predictable way or form a shape when placed in a graph. This type of data pattern is usually described in terms of its features, like centred, spread or shape. Table 13.1 describes these patterns and gives an example of each.

Table 13.1: *Data pattern types*

TYPE	DEFINITION	EXAMPLE
Centred patterns	Pattern where the data has a value in the middle of a series of values arranged from smallest to largest (the median).	
Spread patterns	Pattern where the data varies over a range. If the range is wide, the spread is larger, if the range is small, the spread is smaller.	
Shaped patterns	The data forms a shape when plotted on a chart.	 Skewed right  Skewed left

Activity 13.5

Answer the following questions in your own words.

1. What is data verification?
2. Why do you use data verification?
3. Name four data validation techniques, describe how they work and give an example of each.
4. What are two main types of data entry mistakes?
5. Give three examples of trends and how they would be displayed in a report.
6. When analysing data, what is a "pattern"?
7. Analyse the data from Activity 13.4 and determine trends and patterns if any are present. Present your findings to the class.
8. Write a short paragraph to interpret the graph Petro Price 2017 in Figure 13.6 and derive a conclusion from it.

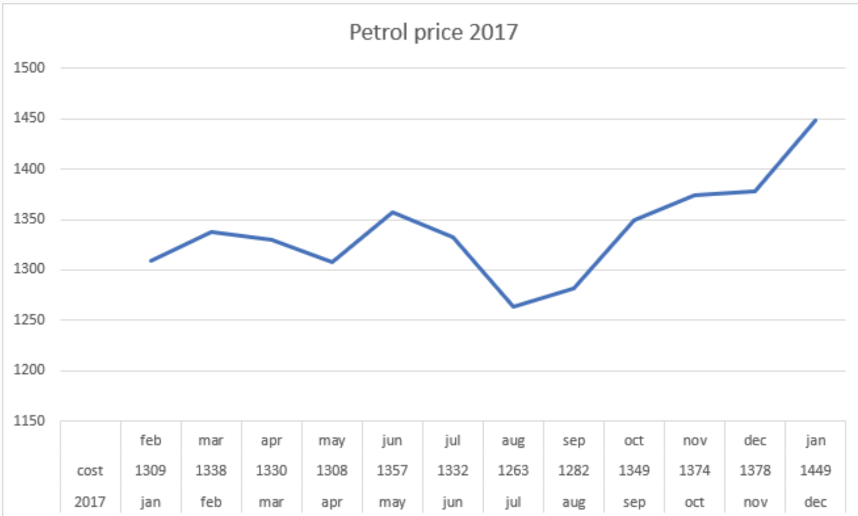
REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

1.1 Data handling can be used to do which of the following? (1)

- **A.** Collect information
- **B.** Organise data
- **C.** Sort information
- **D.** Analyse data

1.2 Which information can be gathered from the following data? (1)



- **A.** The price of petrol in 2018
- **B.** The components of petrol
- **C.** The trend of petrol prices during the year
- **D.** The amount of petrol produced in 2017

1.3 Which of the following is the last step of the information management process? (1)

- **A.** Presenting information
- **B.** Collecting data
- **C.** Processing a database
- **D.** Analysing results

1.4 When evaluating a website, which of the following questions must not be asked? (1)

- **A.** Who supports this website?
- **B.** Who is this website aimed at?
- **C.** Which year was the website established?
- **D.** Who is the author?

1.5 When evaluating information, which of the following must not be determined? (1)

- **A.** Authority
- **B.** Accuracy
- **C.** Objectivity
- **D.** Consistency

QUESTION 2: TRUE OR FALSE

Choose the answer and write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

- **a.** <u>Spreadsheets</u> store their data on a server. (1)
- **b.** <u>Databases</u> are able to perform complex calculations on small amounts of data. (1)
- **c.** <u>Analysis</u> is the first step of the information management process. (1)
- **d.** Surveys and questionnaires can be <u>electronic</u>. (1)
- **e.** If a website does not have the name of the author, it is a <u>good</u> website. (1)

… continued

… continued

QUESTION 3: CATEGORISATION QUESTIONS

3.1 Which of the following situations would use a database or spreadsheet? (3)

SCENARIO	DATABASE/SPREADSHEET
a. Mari wants to organise the music in her playlist from most favourite to least favourite and according to genre.	
b. Agnes wants to use the data from this year's soccer game data to predict who will win the next soccer match.	
c. Martinique wants to keep a record of all the products in her grocery store as well as all her sales and expenses.	

3.2 Which of the following types of graph would best represent the following scenario? (3)

A. Bar graph
B. Line graph
C. Histogram
D. Pie chart

SCENARIO	TYPE OF GRAPH
a. Determining what percentage of the student votes went to each of the four class representatives.	
b. Determining who in the chess club has won the most chess games.	
c. Determining the trend of iPhone sales.	

QUESTION 4: SHORT QUESTIONS

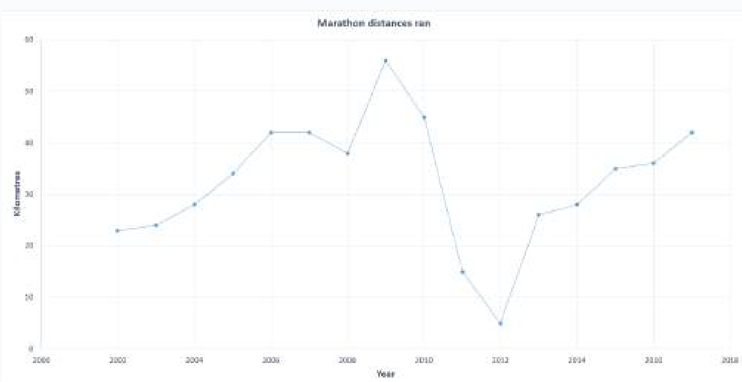
4.1 List two tools you can use to gather data. (2)

4.2 What is the difference between a trend and a pattern? (2)

4.3 Why should you avoid questions that start with "Why"? Also mention how you can prevent this. (2)

QUESTION 5: SCENARIO-BASED QUESTIONS

In order to lose weight and become healthier, Greg decided to start jogging so that he could run in different marathons. Each year Greg recorded how many kilometres he ran. Look at the following graph and answer the questions that follow.



… continued

REVISION ACTIVITY *… continued*

5.1 What type of graph is this? (1)
5.2 Does this graph represent data or information? Give an explanation for your answer. (2)
5.3 According to this graph, when did Greg run the longest distance? (1)
5.4 What is the trend from 2002 to 2009? (1)
5.5 Mention two things that this trend indicates about Greg's health. (2)
5.6 Greg badly injured his ankle during a marathon and was advised by the doctor not to run for the next year. According to this graph, in which year did this injury occur? (1)

Nomfundo had to do research on the different phone brands on sale at the Innovation Tech store. While doing this she found that they sold the following brands:

- Samsung
- Apple
- Nokia
- Blackberry
- Huawei
- LG

Now Nomfundo wants to use this information to figure out what the most popular phone brands are at her school so she can know if it would be a good idea to promote this store at her school.

5.7 Based on this scenario, identify the two problems Nomfundo must solve. (2)
5.8 What information-gathering method Nomfundo can use to solve this problem? Give a reason for why she should use this method, and a reason for why she cannot use the other two methods. (4)
5.9 What two tips would you give Nomfundo about creating questions? (2)
5.10 Create two questions that Nomfundo can ask her school peers to find the information she needs. (2)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU …	YES	NO
1.	Create a task definition?		
2.	Gather data and information using various tools?		
3.	Describe how to evaluate data to ensure that it is reliable and accurate?		
4.	Discuss how to evaluate the validity of a website as a source of information?		
5.	Evaluate information for accuracy and validity?		
6.	Describe the role of spreadsheets and databases in data handling?		
7.	Discuss tools and techniques for data handling and how to extract data?		
8.	Describe how to analyse trends and patterns in data?		

Glossary

A

acceptable use policy (AUP) a document stipulating constraints and practices that a user must agree to for access to a corporate network or the internet

adware any software application that allows advertising banners to be displayed when another program is running

ALS a disease that gradually paralyses people

analogue signal a continuous signal. It is the opposite of a digital signal that sends information in groups

archive refers to materials (such as recordings, documents, or computer files) that have been stored on a system for easy access when needed

arithmetical deals with non-negative real numbers

ascending order from smallest to largest, increasing

auditory refers to hearing

authentication the process of working out whether something or someone is, in fact, who or what they claim to be

automated teller machine (ATM) an electronic banking outlet that allows customers to complete basic transactions without going into the bank

B

bandwidth a measurement of the ability of an electronic communications device or system (such as a computer network) to send and receive information

basic input/output system (BIOS) a ROM chip found on motherboards that allows you to access and set up your computer system at the most basic level

biased having or showing an unfair tendency to believe that some people, ideas, etc., are better than others

binary code a coding system using the binary digits 0 and 1 to represent a letter, digit, or other character in a computer or other electronic device

bit short for binary digit and is a single unit of information that can have a value of either 0 or 1

Bluetooth a specification that describes how devices such as mobile phones, computers, or personal digital assistants can communicate with one another

braille a system of writing for blind people in which letters are represented by raised dots

bring your own device (BYOD) the practice of allowing employees to purchase and use their own computing devices for work instead of the business supplying the device

bug refers to an error, flaw, failure or fault in a computer program or system

C

carpal tunnel syndrome a condition that causes pain and weakness in the wrist, hand, and fingers

Computer Aided Design (CAD) software used by architects, engineers, drafters, artists and others to create precision drawings or technical illustrations

copyright a legal means of protecting an author's work

correlation trend a statistical measure that indicates the extent to which two or more variables fluctuate together

crawl large data-sets developed to browse the WWW in a methodical, automated manner

credentials a piece of information that is sent from one computer to another to check that a user is who they claim to be

cyber-bullying the use of electronic communication to bully a person

cyber-stalking the repeated use of electronic communications to harass or frighten someone

D

data loss prevention policy a strategy for making sure that end users do not send sensitive or critical information outside the corporate network

database an organised collection of data, usually stored in the form of structured fields, tables and columns

defamation the act of damaging a person's reputation

defragmenting the process of reorganising a hard drive's data to help increase the time it takes to run a program and open files

descending order decreasing, from largest to smallest

disk fragmentation when information is deleted from a hard drive and small gaps are left behind to be filled by new data. As new data is saved to the computer, it is placed in these gaps. If the gaps are too small, the remainder of what needs to be saved is stored in remaining gaps

distributed denial of service (DDos) attacks a malicious attempt to interrupt traffic to a server, network or service by flooding it with internet traffic

dots per inch (dpi) a measure of the resolution of an image, both on screen and in print

downtime time during which a computer is out of action or unavailable for use

downward trend refers to a direction from higher to lower

doxing identifying or publishing private information about a person as a form of punishment

E

email spoofing the forgery of an email header so that the message appears to have originated from someone or somewhere other than the actual source

embedded videos videos used within an email for marketing purposes

Energy Star rating the government-backed symbol for energy efficiency

ergonomics a science that deals with designing and arranging things so that people can use them easily and safely

ethernet the standard way to connect computers on a network over a wired connection

ethical refers to doing things that won't do harm to people or the environment

export (data) refers to converting data into another format than the one it is currently in

F

file transfer protocol (FTP) a way to transfer data, particularly files, from one computer to another, usually over the internet but also over a local network

firewall a software program or hardware device that acts as a filter for data entering or leaving a network or computer

format (a disk) refers to preparing a disk to store data

fragmentation when you save a file onto your computer, the computer breaks the file up into smaller pieces in order to store the file on your hard drive

G

geolocation the identification or estimation of the real-world geographic location of an object

gigabit a unit of data size equal to 1 000 megabits

gigahertz (GHz) an indication of the processor's speed. As a general guideline: the higher the frequency, the better the CPU

graphics card a circuit board inside a computer that allows it to receive and show pictures and video

H

high-definition multimedia interface (HDMI) a digital interface used to transmit audio and video data in a single cable

hoax email a scam or trick in an email message, such as warning of a non-existent threat or a request for personal information

I

information and communication technology (ICT) a field of study related to computers and communication networks

import (data) refers to using data produced by another application

information processing cycle the sequence of events in processing information, which includes input, storage, processing, output and communication

installation wizard a feature that takes the user through the steps of installing or the setup of a software program or hardware device

internet protocol (IP) a standard set of rules for sending and receiving data over the internet

intranet a private network for a corporation or organisation that only those with permission can use

L

light emitting diode (LED) a device that lights up and displays information when electricity passes through it

local area network (LAN) a computer network that covers a small area in which the computers in the network share resources, such as internet connections, printers and server connections

live streaming a live transmission of an event over the internet

M

malware malicious software used by cybercriminals

megapixel the resolution of the amount of detail that a camera can capture

memory card a type of storage media

N

near field communication (NFC) a short-range wireless technology that enables simple and secure communication between electronic devices

netiquette refers to internet etiquette. It is a list of rules and guidelines about acceptable behaviour on the internet

network administrator someone responsible for the maintenance and operation of a network or server

non-conductive not able to conduct heat or electricity or sound

O

octa-core (processor) a processor with eight cores

P

passphrase a secret phrase that helps protect accounts, files, folders, and other confidential information

password a string of characters used for authenticating a user on a computer system

pattern refers to data that can also repeat itself in a predictable way or form a shape when placed in a graph

peripheral (equipment) equipment that is connected to a computer but not an essential part of it

phishing a method of trying to gather personal information(such as usernames and passwords) using deceptive emails and websites

pixel refers to any one of the very small dots that together form the picture on a television screen, computer monitor, etc.

pixels per inch (ppi) measures the number of pixels per line per inch in a digital photo

plagiarism the act of copying someone else's work and publishing it as your own

plug-and-play devices devices that can be connected to a computer and used immediately without needing specific driver software

portable refers to something being mobile and easy to carry around

power surge a short, fast rise in voltage that inherently causes an increase in electrical current or vice versa

prevalence something being practised often or over a wide area

Q

quad-core (processor) a processor with four cores

query a request for data or information from a database table or combination of tables

R

random-access memory (RAM) a very fast short-term data storage device that can only store a small amount of information at a time

read-only memory (ROM) stores the basic instructions for what needs to happen when the computer is switched on

really simple syndication (RSS) a way to easily distribute headlines, updates and content to many people at once

repetitive strain injury a painful medical condition that can cause damage to the hands, wrists, upper arms, and backs, especially of people who use computers

resolution the ability of a device to show an image clearly and with a lot of detail

S

salutation a greeting

secure sockets layer (SSL) an encrypted link between you and the server, making the data you and the server send and receive secure and private

social engineering the use of deception to manipulate individuals into giving away confidential or personal information that may be used for fraudulent purposes

sorting the process of arranging data into meaningful order so that you can analyse it more effectively

spam unwanted or irrelevant messages that are sent over the internet or through emails

spyware software installed on a computing device without the user's knowledge that is designed to gather data from that device

structured query language (SQL) a query language used for accessing and modifying information in a database

T

tertiary storage a third level of storage in a computer build to provide huge storage capacity at low cost

tone refers to a shade of colour

trend refers to the general tendency of data to move in one direction over time

U

upward trend means moving in a direction from lower to higher

V

validate the process of comparing data with a set of rules to find out if data is reasonable

verify the process of checking that the data entered exactly matches the original source to find out if data is accurate

video conferencing a live, visual connection between two or more people in separate locations for the purpose of communication

virtual keyboard a computer keyboard that is operated by typing on the screen rather than by pressing physical keys

Voice over Internet Protocol (VoIP) a telephone connection over the internet

volatile memory computer storage that only maintains its data while the device is powered

W

wireless access point (WAP) a wireless receiver that enables a user to connect wirelessly to a network or the internet

website spoofing the act of creating a website as a hoax, with the intention of misleading readers that the website has been created by a different person or organisation

wireless local area network (WLAN) a small network of computers covering a small area, such as a home, office building or school, but with the ability to connect wireless devices such as smartphones, laptops and tablets to the LAN